11.8 and 41.3 kJcm^{-3} (Table 2). These values are 4.6-16 times larger than the vaporization enthalpy of water.

For fs plasmas, in which only one set of free electrons is generated, the free-electron density can be derived from the average energy density (Table 3). It peaks at 520 nm, where it reaches a value of 0.12×10^{22} cm^{-3}. This corresponds to 18% of full ionization of the $1b_1$ valence band level and to a supercritical plasma state (ρ_c/ρ_{crit} = 2.9). At 347 nm, ρ_c reaches 5.2 % of full ionization and the plasma remains subcritical (ρ_c/ρ_{crit} = 0.37), while at 1030 nm, the ionization degree is 7.4 % and the plasma is highly supercritical (ρ_c/ρ_{crit} = 4.8). The complex wavelength dependence reflects the interplay of two counteracting trends: the rate of avalanche ionization increases strongly with λ_L [37] while ρ_{crit} drops with $1/\lambda_L^2$ [Eq. (14)].

The upstream movement of the breakdown wave introduces a spatial inhomogeneity, as can be seen by the colour change in the photographs of IR fs plasma in Fig. 7(a). The change from reddish to bluish hue in upstream direction indicates a shift towards higher average photon energies in the plasma radiation. The spectral composition of the plasma radiation depends on the time-averaged kinetic energy distribution of the CB electrons during emission (as will be discussed in section 5.5.), and a shorter center wavelength corresponds to a larger time-averaged mean energy. The reddish hue in the beam waist region can be explained by shielding from plasma generated during the upstream movement of the breakdown wave, while the bluish hue arises from the longer-lasting energy deposition in the upstream part during the second half of the laser pulse. A colour shift of plasma emission during a movement of the breakdown wave by only 5 µm implicates a sub-micrometer penetration depth of the incident laser light in the plasma. This is consistent with the supercritical electron density at 1030 nm resulting in a very high plasma absorptivity [125,151,236,256]. When ρ_c exceeds ρ_{crit}, both plasma reflectivity and absorptivity increase [151,256], which augments laser-plasma coupling. Above ρ_{crit}, about 30% of the pump light are absorbed within a layer of less than 1 µm thickness, and the rest is reflected or backscattered [236,257]. This way, supercritical electron densities can be reached in bulk material as well as at surfaces.

A comparison of the present results obtained at large *NA* (Table 3) with literature data for smaller focusing angles shows a dramatic increase of plasma energy density, temperature and pressure with increasing *NA* [114,121,122,135,136,208,235,252,258]. Jukna et al. investigated plasma formation in water by weakly focused (*NA* = 0.0035) femtosecond pulses both experimentally and theoretically. They

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showed that an energetic 290-mJ, 800-nm pulse of 500 fs duration produces multiple filaments but no bubble and only weak thermo-elastic acoustic waves. Simulations predicted a peak plasma energy density of 0.030 kJ/cm^3, corresponding to a temperature rise of only 7 K [252]. When 35-µJ, 580-nm pulses of 100-fs duration were focused into water at moderate *NA* = 0.185, evaluation of shadowgraphs and transmission measurements yielded $U_{avg} \approx 1$ kJ/cm^3 [114], slightly above the thermoelastic bubble threshold at 0.6 kJ/cm^3. Correspondingly, the conversion rate into bubble energy remained small (≈ 3 %), even well above threshold [114]. For tighter focusing at *NA* = 0.31, modeling of breakdown in fused silica by 2.5-µJ, 800-nm pulses of 80 fs duration predicted a peak energy density around 2.5 kJ/cm^3 with subcritical electron density ($\rho_c/\rho_{crit} < 0.2$) [136]. The length of the plasma region was here similar to the geometrical length of the laser pulse, which led to some energy depletion by nonlinear absorption during pulse propagation accompanied by pronounced scattering. For breakdown in water and fused silica at *NA* = 0.5 and pulse energies of up to 30 times above breakdown threshold, Potemkin and coworkers measured volumetric energy densities between 4 kJ/cm^3 and 14 kJ/cm^3 but electron densities remained below the critical density [218,235]. The present investigations for breakdown in water at *NA* = 0.8 yield plasma energy density values up to ≈ 40 kJ/cm^3, which for visible and IR wavelengths are associated with supercritical electron densities. The comparison of our results for large and moderate NA with literature data reveals a strong growth of *U* with increasing *NA*, which is due to a decreasing influence of nonlinear propagation effects, a transition from downstream to an upstream movement of the breakdown wave, and a shortening of the plasma length. This picture is consistent with model predictions for breakdown in water at very large *NA* [102,130,131].

What are the upper limits of fs plasma energy density? Since it increases with numerical aperture, energy densities produced by focusing at *NA* > 1 may exceed the values listed in Table 3 for *NA* = 0.8. However, it is unlikely that they will be two orders of magnitude larger as has been stated for focusing of femtosecond pulses into sapphire at *NA* = 1.35 [121,122,259]. In those reports, it was assumed that the plasma zone in bulk sapphire is as thin as the plasma skin layer evolving during breakdown at material surfaces [137,138], namely 65 nm. That assumption led to the conclusion that fs pulses of 20 to 120 nJ energy were absorbed in a volume of only 0.01 µm^3 with an energy density of several MJ/cm^3 and that cavities observed in the target material were produced by shock-wave-induced displacement of *cold* material surrounding the plasma [121,122]. A zone of amorphous material observed around the cavity was attributed to shock-wave-induced modifications of the target crystal, with peak pressure in the order

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of several TPa for a 100-nJ pulse [121,122]. This interpretation is questioned by our measurement results for water that has a band gap of ≈ 9.5 eV [36], close to the values for sapphire and fused silica of ≈ 9 eV [7,38]. The plasma photographs in Fig. 7(a) show luminescent plasma of 4.8 μm^3 volume at E_L = 95 nJ and 11.9 μm^3 volume at E_L = 200 nJ (Fig. 9a), corresponding to an average plasma energy density of ≈ 10 kJ/cm^3, more than hundred times smaller than the extreme state of matter reported in [121,122]. These results for breakdown in water at NA = 0.8 are consistent with modeling results of plasma and cavity formation in fused silica that yielded an energy density of 20 kJ/cm^3 for a 10-nJ, 800-nm, 100-fs pulse focused at NA = 1.35 [215]. With increasing pulse energy, the energy density predicted by the simulations did not increase but remained approximately constant due to the growing volume of the energy deposition zone, in agreement with our experimental observations. The predicted size of the energy deposition zone was ≈ 0.5 μm^3, similar to the size of the amorphous region reported in [121]. This suggests that the cavities in sapphire are formed in a region within the plasma that is softened by nonthermal or thermal melting [136,179,260], and the amorphous zone is due to rapid cooling that prevents re-crystallization. This interpretation is supported by time-resolved investigations of the electron density distribution during void creation in fused silica reported in Refs. [156,261]. Altogether, these results strongly suggest that the plasma energy density in bulk dielectrics at large NA is limited by the movement of the breakdown front during the laser pulse, which shields the beam waist and increases the plasma size for growing E_L [108,114,134,223].

In comparing plasma pressure data for laser-induced breakdown in water and solid dielectrics, one needs to consider the different equations of state of both materials. The pressure produced by a certain energy density is linked to the mass density and sound velocity in the medium [215,262], and both values are much larger in sapphire or fused silica than in water. Therefore, an energy density of 20 kJ/cm^3 leading to cavity formation in fused silica [215] corresponds to a peak pressure of ≈ 250 GPa in SiO_2 but to only 6.5 GPa in water (Fig. 3).

While there is a clear trend towards higher plasma energy density with increasing NA, there is no monotonous increase of U with growing pulse energy at large NA. Above threshold, U first increases rapidly and then stays approximately constant in an energy interval up to about 30 times E_{th} (Fig. 9 and Refs. [218,235]). However, for pulse energies far above the breakdown threshold, U decreases again [114,219,253]. The decrease starts, when the length of the region, in which plasma can be formed at the given pulse energy is longer than the laser pulse. The breakdown wave then changes direction and its

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